Spencer Wallace

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Education

University of Washington Seattle, Washington
PhD Astrophysics *August 2023*

University of Arizona Tucson, Arizona
BS Computer Science, Astronomy and Physics *May 2014*

Experience

Postdoctoral Researcher
University of Washington, Astronomy Department *September 2023 – Present*

- Deployed and optimized the large-scale C++/CUDA simulation code (ChaNGa) on NVIDIA Grace Hopper CPU–GPU nodes at TACC, improving runtime scalability on up to 500+ GPUs and 30,000+ cores
- Profiled and debugged GPU memory and performance bottlenecks with NVIDIA tools, reducing stalls and improving throughput by a factor of 5x
- Collaborated with HPC specialists across physics, CS, and biology to adapt algorithms to diverse workloads

Research Assistant
University of Washington, Astronomy Department *October 2019 – August 2023*

- Designed and implemented a parallel tree-based collision detection module in C++/MPI, enabling the first planet formation simulations with realistic-sized bodies, which resulted in three first-author publications
- Contributed to paratreet, a toolkit for testing/tuning distributed tree traversal algorithms, which resulted in a coauthored IEEE publication
- Built a PyTorch-based generative diffusion model to synthesize simulation results, saving 900k+ CPU hours
- Mentored 5 students in HPC workflows and performance tuning on clusters

Teaching Assistant
University of Washington, Astronomy Department *October 2015 – September 2019*

- Led weekly discussion sections, graded exams, designed homework assignments and performed guest lectures for undergraduate astronomy classes, some of which involved teaching basic data analysis and numerical modeling skills using Python and Excel
- Collaborated with a team of other teaching assistants to ensure exams and assignments were graded consistently and fairly

Software Consultant
LSST Corporation *June 2011 – August 2014*

- Designed interactive visualization widgets (JavaScript/Node.js) for ASCOT the astronomer's collaborative toolkit
- Built a fully customizable, persistent dashboard interface, along with documentation to facilitate large survey data exploration

Skills

Programming Languages: C++, Python, Bash

Tools and Environments: Git, NumPy, pandas, PyTorch, SLURM, matplotlib, seaborn

Communication:3 first-authored publications, 2 co-authored publications, 8 conference talks, 3 conference posters, 20 pop-sci articles published through astrobites